CHAPTER 23

Reflection and Object Space

One of the advantages of dynamic languages such as Ruby is the ability to *introspect*—to examine aspects of a program from within the program itself. This is also called *reflection*.

When people introspect, we think about our thoughts and feelings. This is interesting, because we're using thought to analyze thought. It's the same when programs use introspection—a program can discover the following information about itself:

- What objects it contains
- Its class hierarchy
- The attributes and methods of objects
- Information on methods

Armed with this information, we can look at particular objects and decide which of their methods to call at runtime—even if the class of the object didn't exist when we first wrote the code. We can also start doing clever things, perhaps modifying the program while it's running. We're going to look at Ruby's ObjectSpace, which allows us to reflect on Ruby's internals. Later in this chapter we'll look at distributed Ruby and marshaling, two reflection-based technologies that let us send objects around the world and through time.

Looking at Objects

Have you ever craved the ability to traverse all the living objects in your program? We have! Ruby has a global object called `ObjectSpace` that lets you do some fun tricks with the set of objects Ruby is tracking (this means all the objects that have been created and not yet destroyed by garbage collection).

What Is Garbage Collection?



Ruby is a dynamic language, and it does not require the programmer to manage the memory the program uses during runtime. Instead, Ruby uses a process called garbage collection. Garbage collection looks for objects that have been allocated into memory but are no longer in scope or are otherwise not accessible. Those objects are released back into the memory heap so that the space can be reused.

We're not going to go into the details of how Ruby handles garbage collecting in this book. For one thing, the details change from version to version as newer and better algorithms for identifying discarded objects are developed.